LoanBazaar®

Secure Distribution Venture Pvt. Ltd.

What Is Solar Project Funding ?

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Module 1: Product Overview & Strategic Foundation

1.1 Macroeconomic Necessity & Green Energy Transition Landscape

Solar Project Funding in India represents specialized long-term infrastructure project financing structured specifically to fund the design, procurement, land acquisition, civil construction, equipment installation, grid interconnection, and long-term operation of solar photovoltaic (PV) power generation assets. As India accelerates toward its national target of 500 Gigawatts (GW) of non-fossil fuel energy capacity by 2030 and net-zero carbon emissions by 2070, solar power has emerged as the single largest driver of capital deployment in the clean energy sector.

The transaction architecture of solar project funding differs fundamentally from standard corporate borrowing. In a traditional corporate loan, lenders rely primarily on the corporate sponsor's historical balance sheet, general business cash flows, and overall net worth. Conversely, solar project funding is underwritten on a **limited-recourse or non-recourse basis**, where the debt service capability is evaluated strictly against the future, contractually guaranteed cash flows generated by the standalone solar project itself.

1.2 The Solar Infrastructure Capital Paradox: Fixed Capex vs. 25-Year PPA Inflows

The core financial rationale behind solar project funding addresses the **Solar Infrastructure Capital Paradox**: solar power generation is characterized by exceptionally high upfront capital expenditure (CAPEX) per megawatt, paired with virtually zero fuel costs and minimal, highly predictable operating expenses (OPEX) over a 25-year operational lifespan.

A typical utility-scale solar project in India requires an initial capital outlay ranging from **₹4.00 Crores to ₹5.20 Crores per Megawatt (MW)** (depending on whether tracking systems and Battery Energy Storage Systems are incorporated). This upfront investment must cover: - Tier-1 Solar PV Modules (Bifacial Monocrystalline PERC / N-Type TOPCon panels) - Central or String Inverters and Power Conditioning Units (PCUs) - Solar Mounting Structures (Fixed Tilt or Single-Axis Smart Trackers) - Balance of Plant (BOP) Civil and Electrical Works - High-Voltage Transformers, Switchyards, and Grid Evacuation Transmission Lines - Land Acquisition or Long-Term 30-Year Leasehold Rights

Once commissioned, a solar power plant consumes no fuel (as solar irradiance is free and abundant), resulting in operational maintenance costs of only 1.0% to 1.5% of annual revenue. However, because solar plants sell electricity under 25-year Power Purchase Agreements (PPAs) at regulated or competitively auctioned tariffs (e.g., ₹2.50 to ₹3.20 per kWh), generating a modest annual cash margin, a developer cannot amortize a ₹500 Crore capital investment over a short 3-to-5 year corporate loan tenure without causing an immediate cash flow default.

Solar project funding resolves this paradox by providing **long-tenure, sculpted debt financing (15 to 20 years)** structured to align perfectly with the 25-year PPA lifespan, incorporating an initial construction grace period (moratorium) during which interest is capitalized, thereby ensuring the project achieves robust Debt Service Coverage Ratios throughout its operational cycle.

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1.3 Classification of Solar Asset Classes in India

Solar project funding in India spans five distinct commercial asset categories, each possessing unique risk-return profiles, off-taker credit profiles, and legal structures:

1. **Utility-Scale Grid-Connected Solar IPP Projects:** Large-scale ground-mounted solar farms (ranging from 50 MW to 1,000+ MW) connected directly to the Inter-State Transmission System (ISTS) or Intra-State Transmission System (InSTS). Power is sold under 25-year fixed-tariff PPAs to sovereign-backed central aggregators such as Solar Energy Corporation of India (SECI), NTPC Limited, or NHPC, or directly to State Distribution Companies (DISCOMs).
2. **Commercial & Industrial (C&I) Open Access Solar Projects:** Ground-mounted or large rooftop solar projects installed to supply clean power directly to commercial and industrial consumers (e.g., automobile OEMs, data centers, textile mills, pharmaceutical plants) across the state grid under Open Access regulations. Off-take is governed by 10-to-15 year bilateral PPAs executed at tariffs discounted below DISCOM retail grid rates.
3. **Group Captive Open Access Solar Projects:** Highly popular structured C&I solar framework established under Section 9 of the Electricity Act 2003 and Rule 3 of the Electricity Rules 2005. The industrial consumer holds a mandatory **26% equity shareholding** in the solar SPV and contracts to consume at least **51% of the power generated**. This structure legally exempts the off-taker from paying heavy DISCOM Cross-Subsidy Surcharges (CSS) and Additional Surcharges (AS), unlocking tariff savings of ₹1.50 to ₹2.50 per unit.
4. **Floating Solar Photovoltaic (FPV) Projects:** Solar PV arrays mounted on floating structures deployed across water bodies such as hydroelectric reservoirs, industrial lagoons, and lakes. FPV eliminates land acquisition costs, reduces water evaporation, and benefits from higher panel generation efficiency due to the natural cooling effect of water.
5. **Hybrid Solar-Wind with Battery Energy Storage Systems (BESS):** Advanced multi-technology clean energy plants combining solar PV arrays, wind turbine generators, and utility-scale lithium-ion/flow battery storage. Structured to deliver Firm and Dispatchable Renewable Energy (FDRE) or Round-The-Clock (RTC) power to meet peak grid demand.

1.4 Regulatory, Statutory & Policy Anchors

Solar project funding operates within a highly regulated legal and statutory environment in India:

- **The Electricity Act, 2003:** The foundational legislation governing power generation, transmission, distribution, and open access trading across India.
- **Green Energy Open Access Rules, 2022:** Regulatory reform lowering the open access eligibility limit from 1 Megawatt down to **100 Kilowatts (kW)**, mandating time-bound open access approval within 15 days, and rationalizing cross-subsidy and banking charges.
- **MNRE Approved List of Models and Manufacturers (ALMM):** Mandatory mandate enforced by the Ministry of New and Renewable Energy requiring all government-backed, SECI, DISCOM, and

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open access solar projects to procure PV modules strictly from ALMM-listed domestic manufacturing facilities.

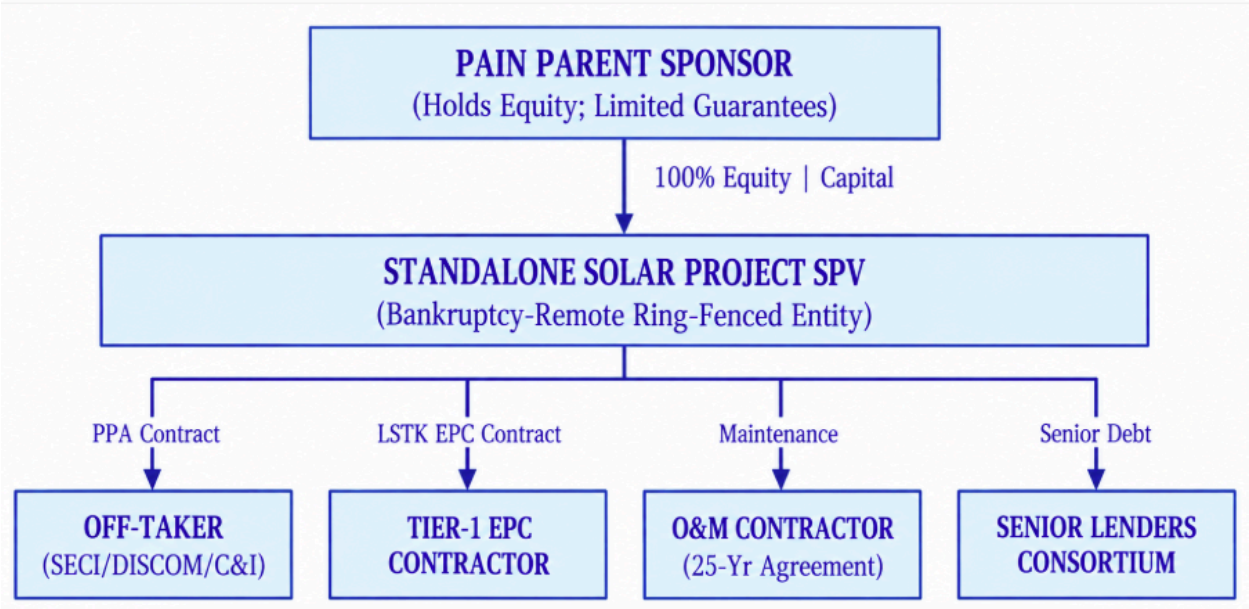
- **Reserve Bank of India (Project Finance) Directions 2025/2026:** Regulatory framework governing bank lending to project finance SPVs, enforcing strict 5.0% standard asset provisioning during the construction phase, DCCO monitoring, and mandatory minimum Debt Service Coverage Ratios.
- **Section 58(f) of the Transfer of Property Act, 1882:** Statutory provision governing the creation of Equitable Mortgages over solar park land and leasehold rights via the deposit of title deeds.
- **Section 77 of the Companies Act, 2013:** Mandates the registration of exclusive corporate charges over SPV assets with the Registrar of Companies (ROC Form CHG-1) within 30 days of loan execution.
- **SARFAESI Act, 2002 & CERSAI:** Grants lenders statutory power to enforce security interests, seize solar plant assets, and take over project operations upon default without mandatory court intervention, registered on the Central Registry of Securitisation Asset Reconstruction and Security Interest (CERSAI) portal.

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Module 2: Structural Architecture & How Solar Project Funding Works

2.1 The Bankruptcy-Remote Non-Recourse / Limited-Recourse SPV Model

The structural cornerstone of Solar Project Funding in India is the **bankruptcy-remote Special Purpose Vehicle (SPV)**. The parent sponsor company incorporates a dedicated, standalone ring-fenced corporate SPV (a Private Limited or Public Limited company) whose sole business mandate is to hold title to the project land, execute project contracts, construct, own, and operate the solar power plant.



By isolating the solar project inside an independent SPV: 1. **Lender Legal Isolation:** Lenders' debt recourse is limited exclusively to the assets, contracts, cash flows, and accounts of the SPV. The liabilities or operational failures of the parent sponsor cannot drag the solar project into insolvency. 2. **Sponsor Shielding:** Conversely, if another subsidiary of the parent sponsor defaults, creditors of that defaulting entity cannot attach the cash flows or solar PV assets of the bankruptcy-remote solar SPV. 3. **High Financial Leverage:** Because the project cash flows are contractually locked under long-term PPAs, lenders allow the SPV to raise higher debt leverage (**70:30 to 80:20 Debt-to-Equity ratio**) than a standard corporate balance sheet could ever support.

2.2 Off-take Models & Contractual Power Purchase Agreements (PPAs)

The bankability and debt capacity of a solar project depend entirely on the legal robustness, credit rating, and contractual enforceability of its Power Purchase Agreement (PPA). Three dominant PPA frameworks exist in India:

1. Sovereign-Backed Central Utility PPAs (SECI / NTPC / NHPC)

- **Tenure:** 25 Years Fixed.

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- **Credit Profile:** AAA Sovereign Equivalent.
- **Payment Security Mechanism (PSM):** Includes a mandatory 3-month Letter of Credit (LC) opened by SECI in favor of the SPV, backed by a Payment Security Fund, and supported by a Tripartite Agreement (TPA) between the Ministry of Power, Reserve Bank of India, and State Governments allowing direct deduction from State RBI allocation funds in case of DISCOM payment defaults.
- **Bankability:** Highest. Attracts the lowest interest rate spreads (8.50% to 9.25% p.a.) and longest amortization tenures (18 to 20 years).

2. Group Captive Open Access PPAs

- **Tenure:** 10 to 15 Years.
- **Credit Profile:** AA or A-rated Commercial & Industrial Corporations.
- **Structural Mandate:** Off-taker owns 26% equity in the SPV and consumes $\geq 51\%$ of generated electricity under Electricity Rules 2005.
- **Exemption:** Statutory exemption from Cross-Subsidy Surcharge (CSS) and Additional Surcharge (AS).
- **Bankability:** High. Requires stringent off-taker credit appraisal, termination buyout clauses, and Minimum Guaranteed Off-take (Take-or-Pay) covenants.

3. State DISCOM PPAs

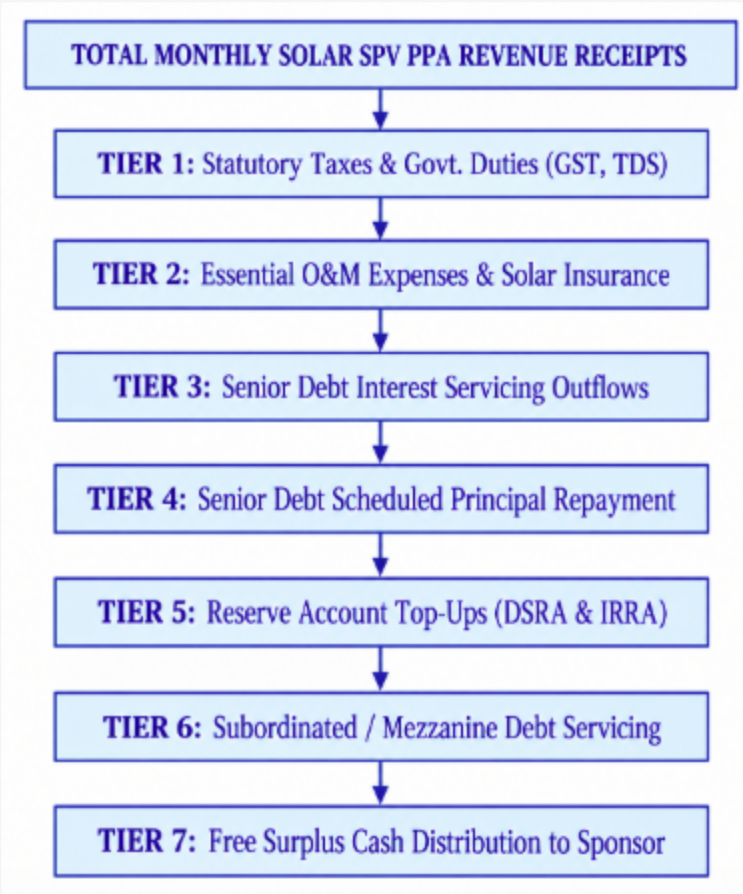
- **Tenure:** 25 Years.
- **Credit Profile:** A to BB-rated State Electricity Distribution Companies.
- **Key Risks:** Delayed monthly tariff realization (often 60 to 180 days delayed), grid curtailment risk, and state regulatory renegotiation threats.
- **Bankability:** Moderate. Lenders mandate higher Debt Service Reserve Accounts (2 Quarters DSRA) and higher interest rate spreads (9.75% to 11.00% p.a.).

2.3 Solar Project Cash Flow Ring-Fencing & 7-Tier TRA Escrow Waterfall

To eliminate cash leakage, all project revenues—including monthly PPA energy payments from SECI, DISCOMs, or C&I off-takers, CDM carbon credit proceeds, and insurance claims—must be deposited directly into a bank-controlled **Trust and Retention Account (TRA) Escrow Account**.

The TRA Escrow Agent operates the account under a legally binding Tripartite Agreement between the SPV, the Senior Lenders, and the Escrow Bank. Monthly disbursements are governed strictly by a **7-Tier Cash Flow Priority Waterfall**:

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Module 3: Comparative Capital Architecture Matrix

3.1 Structural Comparison Across Solar Financing Instruments

Parameter	Utility-Scale SECI PPA Solar Debt	C&I Group Captive Open Access Debt	C&I Rooftop Solar Capex Debt	Green Bond / InvIT Post-COD Refinancing	Private Credit / Mezzanine AIF Debt
Primary Borrower Entity	Standalone Bankruptcy-Remote SPV	Ring-fenced Group Captive SPV	Operating Corporate / MSME Parent	Operational Asset SPV / InvIT Trust	Parent Sponsor / HoldCo SPV
Off-taker Risk Profile	Sovereign / SECI (AAA Equivalent)	AA to A-rated C&I Corporate	Self-Consumption Corporate / MSME	Multi-Asset Diversified Pool	Parent Corporate Risk
Max Permissible Debt-to-Equity	75:25 to 80:20	70:30 to 75:25	65:35 to 70:30	80:20 to 85:15	N/A (Subordinated / Junior Debt)
Indicative Interest Benchmark	EBLR / MCLR + 1.25% to 2.00%	EBLR / MCLR + 2.00% to 3.00%	EBLR / MCLR + 2.50% to 4.00%	Fixed Coupon NCDs (7.50% to 8.75%)	Fixed Coupon (14.00% to 18.00% p.a.)
All-In Effective Yield Range	8.50% to 9.25% p.a.	9.25% to 10.50% p.a.	9.75% to 11.50% p.a.	7.75% to 8.75% p.a.	14.50% to 18.50% p.a. (Gross IRR)
Construction Moratorium	12 to 18 Months (Pre-COD)	6 to 12 Months	3 to 6 Months	Zero (Fully Operational Asset)	12 to 24 Months
Door-to-Door Amortization	15 to 20 Years Post-COD	10 to 14 Years	5 to 8 Years	10 to 15 Years	3 to 5 Years
Primary Security	Mortgage on Land + Hypothecation + PPA Assignment	Mortgage on Land + PPA Assignment + 26% Equity Pledge	Hypothecation of Rooftop Equipment + LAP	Exclusive First Mortgage + TRA Escrow Charge	Second Charge + Subordinated Share Pledge
Minimum Required DSCR	1.20x to 1.25x	1.25x to 1.30x	1.30x to 1.40x	1.15x to 1.20x	1.10x to 1.15x

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Parameter	Utility-Scale SECI PPA Solar Debt	C&I Group Captive Open Access Debt	C&I Rooftop Solar Capex Debt	Green Bond / InvIT Post-COD Refinancing	Private Credit / Mezzanine AIF Debt
DSRA Buffer Requirement	2 Quarters Principal + Interest	2 Quarters Principal + Interest	1 Quarter EMI	1 Quarter EMI	N/A
Section 32 Acc. Dep. Eligibility	Applicable to SPV	Applicable to SPV / C&I Equity	Applicable to Parent Company	N/A	N/A
Section 36(1)(iii) Tax Shield	Fully Deductible (SPV P&L)	Fully Deductible (SPV P&L)	Fully Deductible (Parent P&L)	Fully Deductible	Fully Deductible

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Module 4: Common Eligibility Criteria, Credit Appraisal & Underwriting Framework

4.1 Solar Resource Assessment & P50/P90/P99 Generation Probability

Unlike conventional thermal or hydro power plants, a solar power plant's revenue generation capacity is governed by atmospheric solar irradiation, which varies dynamically across daily and seasonal cycles. Lenders require a independent **Solar Resource Assessment (SRA)** report authored by certified global solar engineering consultants (e.g., DNV GL, TÜV Rheinland, Mott MacDonald, Lahmeyer, or Black & Veatch).

The SRA report utilizes 20+ years of satellite solar irradiance data (Meteonorm, SolarGIS, NASA) combined with ground-based On-Site Weather Monitoring Station (AWS) pyranometer measurements to model three statistical generation probability metrics:

- **P50 Generation Estimate:** The statistical median level of electricity generation expected to be equaled or exceeded **50% of the time** over a 1-year period. P50 represents the baseline expected mean generation.
- **P90 Generation Estimate:** A conservative, risk-adjusted generation figure expected to be equaled or exceeded **90% of the time**. P90 accounts for statistical weather variations, cloud cover, and lower irradiation years.
- **P99 Generation Estimate:** An extreme downside stress-scenario generation figure expected to be equaled or exceeded **99% of the time**.

The Golden Underwriting Rule of Solar Debt Sizing:

Institutional lenders and bank credit committees in India **never size project debt or schedule debt service against P50 estimates**. Debt sizing and repayment schedules are calculated strictly against **P90 generation probability estimates**. Sizing debt against P90 guarantees that even during cloudier, lower-than-average solar irradiation years, the project's cash flow will comfortably cover 100% of senior debt service obligations.

4.2 Technical Performance Ratios: CUF, PR & Module Degradation Dynamics

When evaluating a solar project credit proposal, lenders examine three core technical performance metrics:

1. Capacity Utilization Factor (CUF)

CUF represents the ratio of actual annual energy generated (in kWh) relative to the theoretical maximum output if the plant operated at 100% rated peak capacity 24 hours a day, 365 days a year:

$$extCUF(\%) = (racextAnnualEnergyGeneration(kWh)extInstalledDCCapacity(kWp)imes8,760extHoursight)imes100$$

In India, utility-scale solar plants achieve CUF benchmarks ranging from **21.0% to 28.0%**, depending on solar irradiance in the location (e.g., Rajasthan and Gujarat achieve 25%–28% CUF; Central/Southern

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states achieve 21%–24% CUF). The integration of **Single-Axis Solar Trackers** (which follow the sun's path east to west) increases CUF by **3.5% to 5.0%**, while **Bifacial Modules** (which absorb reflected light from the ground albedo on their rear side) add another **2.0% to 3.5%** CUF gain.

2. Performance Ratio (PR)

PR measures the overall quality and efficiency of the solar plant, accounting for temperature losses, inverter conversion losses, DC/AC cable losses, transformer losses, and module field mismatch:

$$extPerformanceRatio(PR\%) = (racextActualACEnergyDeliveredtoGrid(kWh)extCalculatedTheoret$$

A well-designed, Tier-1 utility solar plant must achieve a baseline **PR of 78.0% to 82.0%** at Commissioning / Commercial Operation Date (COD).

3. Linear Module Degradation Rate

Solar PV panels experience gradual physical degradation over time due to UV exposure and thermal stress. Tier-1 N-Type TOPCon and Mono-PERC modules are backed by 25-to-30 year linear performance warranties guaranteeing: - **Year 1 Initial Degradation:** Maximum 1.0% to 1.5% (including Light Induced Degradation - LID). - **Annual Linear Degradation (Years 2 to 25):** Maximum 0.40% to 0.55% per annum. - **Year 25 Guaranteed Output:** At least **83.0% to 87.5%** of original nameplate capacity.

Lenders' financial models embed this annual 0.5% generation drop into the 20-year cash flow forecast, ensuring debt service coverage ratios remain satisfied even in Year 15 and Year 20 of operations.

4.3 Quantitative Credit Appraisal & Debt Sizing Ratios (CFADS, DSCR, LLCR, PLCR)

Credit committees utilize four core quantitative cash flow coverage metrics to size senior solar project debt:

1. Cash Flow Available for Debt Service (CFADS)

$$extCFADS_t = extEBITDA_t + extInterestIncome_t - extStatutoryTaxes_t - extContractualO\&MOutflows_t - \Delta extNWC_t$$

2. Debt Service Coverage Ratio (DSCR)

$$extDSCR_t = racextCFADS_textSeniorDebtService_t = racextCFADS_textScheduledInterestPayment_t + extScheduledPrincipalRepaym$$

- **Target Average DSCR:** Must equal $1.25xextto1.35x$ over the entire 15-to-20 year loan tenure.
- **Minimum Annual DSCR Floor:** Must not drop below $1.15x$ in any single operational year.

3. Loan Life Coverage Ratio (LLCR)

LLCR calculates the net present value (NPV) of CFADS over the remaining tenure of the debt relative to the outstanding principal debt balance (D_t):

$$extLLCR_t = rac\sum_{i=t}^{N} racextCFADS_i\left(1 + K_d\right)^i D_t \geq 1.35x$$

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Where K_d is the weighted average cost of senior debt and N is the final debt maturity year.

4. Project Life Coverage Ratio (PLCR)

PLCR calculates the NPV of CFADS over the entire 25-year operational lifespan of the solar asset (including the post-debt PPA tail period) relative to the outstanding principal debt balance:

$$extPLCR_t = rac \sum_{i=t}^{T_{PPA}} racextCFADS_i \left(1 + K_d\right)^i D_t \geq 1.50x$$

A high PLCR ($\geq 1.50x$) provides comfort to lenders that even if a debt refinancing or extension becomes necessary, the asset retains immense equity value during its PPA tail period.

4.4 Sponsor Track Record, Credit Ratings & Technical Certifications

In addition to quantitative project cash flow metrics, credit appraisal mandates strict eligibility filters for the parent sponsor, EPC contractor, and equipment OEMs:

- **Promoter Credit Rating:** Minimum corporate credit rating of **CRISIL / ICRA / CARE BBB-** (Investment Grade) for the parent sponsor, with a CIBIL Commercial Rank of **CMR 1 to CMR 4**.
- **Promoter Execution Track Record:** Proven track record of successfully constructing and operating at least **100 MW** of utility or C&I solar capacity in India over the preceding 3 to 5 years.
- **MNRE ALMM & BIS Compliance:** Solar PV modules must be strictly sourced from Tier-1 manufacturers listed on the **MNRE Approved List of Models and Manufacturers (ALMM)** and possess valid Bureau of Indian Standards (BIS) IS 14286 / IEC 61215 certifications.
- **Tier-1 BloombergNEF Module Manufacturer List:** Modules must be procured from manufacturers listed on the BloombergNEF (BNEF) Tier-1 PV Module Maker list for at least 4 consecutive quarters.

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Module 5: Security Architecture, Documentation & Legal Covenants

5.1 Primary Collateral & Equitable Mortgage Structuring (Sec 58(f) Transfer of Property Act)

Senior solar project debt is secured by a multi-layered, first-ranking collateral package created in favor of the Security Trustee (acting on behalf of the consortium of senior lenders):

1. **First Exclusive Equitable Mortgage over Project Land:** First ranking equitable mortgage created under **Section 58(f) of the Transfer of Property Act, 1882** via the deposit of original title deeds / 30-year leasehold agreements covering 100% of the solar park land parcel.
2. **First Exclusive Mortgage over Civil & Civil Structures:** Mortgage covering all control room buildings, inverter stations, switchyards, perimeter fencing, internal access roads, and drainage networks.

5.2 Hypothecation over Movable Plant, Equipment & Receivables (Sec 2(1)(n) SARFAESI Act)

1. **First Exclusive Hypothecation Charge over Movable Plant & Machinery:** First ranking hypothecation charge created under **Section 2(1)(n) of the SARFAESI Act, 2002** over all solar PV modules, mounting structures, tracking systems, central/string inverters, transformers, SCADA monitoring systems, cables, weather stations, and spare inventory.
2. **First Exclusive Hypothecation Charge over Receivables & Bank Accounts:** First charge over 100% of present and future PPA receivables, energy billing invoices, insurance proceeds, carbon credit revenues, TRA Escrow Accounts, Debt Service Reserve Accounts (DSRA), and all operating bank accounts.

5.3 Contractual Assignments & Payment Security Mechanisms (PSM)

1. **Assignment of Project Contracts:** Absolute legal assignment in favor of lenders over:
   - 25-Year Power Purchase Agreement (PPA) with SECI / DISCOM / C&I Off-taker
   - Lump-Sum Turnkey (LSTK) EPC Contract (including Liquidated Damages LD guarantees)
   - 25-Year Operation & Maintenance (O&M) Agreement
   - Transmission Interconnection & Grid Connectivity Agreements with STU/POWERGRID
   - Comprehensive All-Risk & Loss of Profits Insurance Policies
2. **Tripartite Substitution Agreement:** Legally binding agreement executed between the Off-taker (SECI/DISCOM), the SPV, and the Senior Lenders granting lenders the **irrevocable right to step in and substitute the promoter/SPV** with a new qualified operator in the event of project default.
3. **PPA Payment Security Mechanism (PSM):** Irrevocable Revolving Letter of Credit (LC) opened by SECI or the DISCOM in favor of the SPV equivalent to **1 to 3 months of peak monthly power billing**.

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5.4 Statutory Charge Registrations (CERSAI, ROC Form CHG-1, NeSL IU)

To enforce security validity against third parties and liquidators, senior lenders mandate three mandatory statutory filings: - **ROC Form CHG-1 Filing:** Mandatory registration of the corporate charge over SPV assets with the Registrar of Companies under **Section 77 of the Companies Act, 2013** within **30 days** of document execution. - **CERSAI Portal Registration:** Mandatory registration of the mortgage and hypothecation charge on the Central Registry of Securitisation Asset Reconstruction and Security Interest of India portal within 30 days. - **NeSL Information Utility Registration:** Recording of the debt facility and financial default covenants on the National E-Governance Services Limited (NeSL) portal under the Insolvency and Bankruptcy Code, 2016.

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Module 6: Indicative Tenure, Pricing Mechanics, Accounting & Tax Shields

6.1 Pricing Benchmarks & Spread Sizing Mechanics

Solar Project Funding in India is priced dynamically based on benchmark floating reference rates (Repo Rate / EBLR or 1-Year MCLR) plus a project-specific credit spread reflecting off-taker risk, location, and technology:

$$extNominalInterestRate(\%p.a.) = extRBIRepoRate/MCLRBenchmark + extCreditSpread$$

Benchmark Spreads across Off-take Categories:

1. **Utility-Scale SECI / NTPC Solar Projects:** Repo Rate / MCLR + **1.75% to 2.25%** → Effective Rate: **8.50% to 9.25% p.a.**
2. **C&I Group Captive Open Access Solar Projects:** Repo Rate / MCLR + **2.25% to 3.25%** → Effective Rate: **9.25% to 10.50% p.a.**
3. **C&I Rooftop Solar Projects:** Repo Rate / MCLR + **3.00% to 4.25%** → Effective Rate: **9.75% to 11.50% p.a.**

Lenders incorporate **post-COD interest rate reset clauses** every 1 to 3 years. Upon achieving Commercial Operation Date (COD) and establishing a 12-month track record of stable generation exceeding P90 estimates, lenders reduce the credit spread by **25 to 50 basis points (0.25% to 0.50%)**.

6.2 Door-to-Door Tenures, Moratoriums & Sculpted Amortization Schedules

- **Construction Moratorium Period:** Lenders grant a principal repayment grace period covering the entire construction timeline (**12 to 18 months**) plus a **6-month post-COD buffer**. During construction, monthly interest accrued is capitalized into Capital Work-in-Progress (CWIP) or paid via a pre-funded Interest During Construction (IDC) facility.
- **Door-to-Door Amortization Tenure:** Senior debt is amortized over **15 to 20 years** post-COD. This leaves a mandatory **3-to-5 year PPA Tail Period** (as PPAs run for 25 years), providing a buffer for lenders if amortization needs to be extended due to macro disruptions.
- **Sculpted Amortization Schedule:** Principal repayment installments are not structured as flat equal monthly installments (EMI). Repayments are **sculpted seasonally and annually** to match solar irradiance cycles (e.g., lower repayments in monsoon months July–September, higher repayments in peak summer months March–June) and to account for the 0.5% annual panel degradation.

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6.3 Section 32 Accelerated Depreciation Tax Shield Analysis

One of the most potent corporate tax incentives available under the Indian Income Tax Act, 1961 for solar energy investments is **Accelerated Depreciation (AD)** under Section 32.

- **Tax Depreciation Rate:** Solar energy power plants (PV modules, mounting structures, inverters) qualify for a special Written Down Value (WDV) tax depreciation rate of **40% per annum** under Rule 5 of the Income Tax Rules, 1962.
- **First-Year Half-Year Rule:** If the solar plant is commissioned and put to commercial use for more than 180 days in a financial year (prior to October 3rd), the full 40% depreciation is claimed in Year 1. If put to use for less than 180 days, 20% depreciation is claimed in Year 1, and the remaining 20% is carried forward.

Mathematical Mechanics of Accelerated Depreciation Tax Shield:

For a corporate sponsor investing ₹100 Crores in a Group Captive or Captive Solar plant generating taxable profits in its core business (taxed at the corporate tax rate of $25.168\% = 22\%extbase + 10\%extsurcharge + 4\%extcess$):

$$extYear1WDVTaxDepreciation = ₹100.00extCroresimes40\% = ₹40.00extCrores$$

$$extYear1DirectCorporateTaxCashShield = ₹40.00extCroresimes25.168\% = ₹10.067extCrores$$

Over the first 3 years, Section 32 allows the sponsor to write off over **78.4% of the total plant cost** against corporate taxable profits, generating an immediate cash tax shield equivalent to **~20% of the total project cost**, thereby massively accelerating equity payback and expanding project internal rates of return (IRR).

6.4 Section 36(1)(iii) Interest Tax Shields & Ind AS Accounting Rules

Under **Section 36(1)(iii) of the Income Tax Act, 1961**, 100% of the interest paid on senior project debt utilized for solar plant construction is fully deductible as an operational business expense once commercial generation commences.

$$extEffectivePost - TaxDebtYield(\%p.a.) = extNominalInterestRate(\%)imes\left(1 - T_{extcorp}\right)$$

For a solar SPV borrowing at a nominal interest rate of **8.75% p.a.** subject to the corporate tax rate of **25.168%**:

$$extEffectivePost - TaxDebtCost = 8.75\%imes(1 - 0.25168) = 6.548\%extp.a.$$

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Applicable Indian Accounting Standards (Ind AS):

- **Ind AS 16 (Property, Plant and Equipment):** Mandates component accounting for solar PV modules, inverters, and transformers, depreciating each component over its specific useful life (modules over 25 years; inverters over 10 to 12 years).
- **Ind AS 23 (Borrowing Costs):** Capitalizes interest costs incurred during the 12-to-18 month construction phase directly into Capital Work-in-Progress (CWIP) on the balance sheet.
- **Ind AS 115 (Revenue from Contracts with Customers):** Governs revenue recognition based on monthly certified Joint Meter Reading (JMR) energy generation bills delivered to grid sub-stations.

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Module 7: Suitable Sectors, Borrower Profiles & Market Alignment

7.1 Target Industries, Business Profiles & Off-taker Ecosystems

Solar Project Funding is tailored for five core commercial and industrial segments in India:

1. **Independent Power Producers (IPPs):** Utility-scale clean energy developers (e.g., Renew Power, Avaada, Tata Power Renewables, Adani Green, Hero Future Energies) constructing 100 MW to 1,000+ MW solar parks under SECI/NTPC central tenders.
2. **C&I Heavy Industrial Off-takers (Group Captive):** Energy-intensive manufacturing corporations—such as Steel Plants, Automobile OEMs, Chemical Refineries, Cement Plants, Data Centers, and Textile Mills—seeking to offset 50% to 100% of their thermal power consumption with cheap Group Captive solar power.
3. **Commercial Real Estate & Institutional Facilities:** IT Parks, Shopping Malls, Airport Terminals, Hospitals, and Universities installing captive rooftop or ground-mounted solar plants to lower operational Common Area Maintenance (CAM) electricity charges.
4. **Agricultural Feeder Solarization (PM-KUSUM Scheme):** Agricultural solar project developers constructing 1 MW to 10 MW decentralized solar plants at rural sub-stations under Component A/B/C of the PM-KUSUM scheme to supply free daytime solar power for farm irrigation pumps.
5. **Turnkey Solar EPC Contractors:** Engineering firms requiring non-fund based Working Capital lines (Performance BGs and Inland LCs) to execute mega utility solar contracts.

7.2 Indian Financial Market Credit Footprint Metrics

According to institutional credit market reports from SIDBI, CRISIL, and the Reserve Bank of India: - **Total Commercial Credit Deployment in Renewable Energy:** Exceeds **₹4.50 Lakh Crores** as of FY2026, making renewable energy the single fastest-growing sub-segment of infrastructure debt in India. - **Primary Institutional Lenders:** Specialised Government Financial Institutions (IREDA, Power Finance Corporation - PFC, REC Limited), State Bank of India (SBI - which manages dedicated ESG/Solar debt lines exceeding ₹50,000 Crores), major private commercial banks (HDFC Bank, ICICI Bank, Axis Bank), and international Development Finance Institutions (IFC, ADB, KfW). - **Digital Supply Chain & Vendor Financing Integration:** SBI's digital supply chain platforms (**e-DFS at ₹49,261 Crores** and **e-VFS at ₹28,092 Crores**) provide seamless digital working capital and bill discounting to Tier-1 solar EPC contractors and module suppliers.

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Module 8: Key Advantages and Strategic Limitations of Solar Project Funding

8.1 Core Strategic Advantages for Developers & Corporate Off-takers

1. **Non-Recourse SPV Shielding:** Protects the parent sponsor's balance sheet by isolating debt liabilities strictly inside the bankruptcy-remote solar SPV.
2. **Long Door-to-Door Tenures (15 to 20 Years):** Matches the long 25-year operational life of solar PV assets, lowering annual debt service installments and maximizing project DSCR.
3. **Low All-in Borrowing Cost (8.50% to 9.25% p.a.):** Backed by AAA sovereign-equivalent SECI PPAs or credit-worthy C&I group captive off-takers, solar project debt commands the lowest interest spreads in the infrastructure sector.
4. **Massive Tax Shields:** Combining Section 32 Accelerated Depreciation (40% WDV) with Section 36(1)(iii) Interest Tax Shields slashes effective post-tax borrowing yield to **~6.55% p.a.** and recovers ~20% of project cost in tax cash savings within 36 months.
5. **Significant Tariff Savings for C&I Off-takers:** Group Captive open access solar delivers clean power at **₹3.50 to ₹4.50 per unit**, compared to retail DISCOM grid tariffs of **₹8.00 to ₹11.00 per unit**, unlocking 40% to 60% savings in electricity costs.

8.2 Structural Risks, Operational Bottlenecks & Institutional Risk Mitigants

1. **Solar Irradiation & Weather Intermittency Risk:** Extended monsoons, cloud cover, or dust storms lower annual energy generation below P50 estimates.
   - *Lender Risk Mitigant:* Senior debt sizing is calculated strictly against conservative **P90 generation probability estimates**, supported by a pre-funded 2-Quarter DSRA buffer.
2. **Grid Curtailment & Transmission Congestion Risk:** State transmission utilities (STUs) may order solar plants to curtail power generation during low-demand periods without paying compensation.
   - *Lender Risk Mitigant:* Lenders mandate LTA/GNA (General Network Access) connectivity approvals prior to initial loan disbursal and prioritize projects connected to ISTS national grids.
3. **Module Quality & Technology Degradation Risk:** Substandard solar modules experience rapid Light Induced Degradation (LID) exceeding 1.0% per annum.
   - *Lender Risk Mitigant:* Procurement is restricted strictly to MNRE ALMM-approved domestic manufacturers and BloombergNEF Tier-1 PV module makers backed by 25-year linear performance warranties.

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4. **Off-taker Payment Delay & DISCOM Financial Stress:** State DISCOMs may delay monthly tariff remittances by 90 to 180 days.
    - *Lender Risk Mitigant:* Utility PPAs mandate SECI Payment Security Mechanisms (Revolving LCs & Payment Security Funds) and enforce direct 7-tier TRA Escrow Waterfalls.

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Module 9: Comprehensive Real-World Case Study (Indian Rupee Models)

9.1 Baseline Background & Transaction Parameters

To demonstrate the quantitative mechanics, debt sizing, and tax shield optimization of Solar Project Funding, consider a greenfield **100 Megawatt (MW) AC / 140 MW DC Utility-Scale Grid-Connected Solar PV Project** developed in the Bhadla Solar Park, Rajasthan by an Independent Power Producer: **Mumbai Solar Infra & Power SPV Ltd. (MSIPSL)**, a 100% subsidiary of a corporate clean energy group.

BHADLA 100 MW AC SOLAR PV PROJECT FINANCIAL MODEL	
Total Project Cost Outlay (TPC):	**₹500.00 Crores (100.0%)**
- Tier-1 Bifacial PV Modules (140 MWp DC)	₹280.00 Crores (56.0%)
- Inverters, Trackers & Electrical BOP	₹80.00 Crores (16.0%)
- Land Acquisition & Civil Works (500 Acres)	₹50.00 Crores (10.0%)
- 220 kV Switchyard & ISTS Grid Line	₹40.00 Crores (8.0%)
- Pre-Operative Expenses & Interest (IDC)	₹50.00 Crores (10.0%)
PPA Contract Terms:	25-Year Fixed Tariff @ ₹2.50 / kWh with SECI
Annual Energy Generation (P90 Estimate):	219.00 Million Units (kWh) [CUF = 25.0%]
Gross Annual Revenue Realization:	₹54.75 Crores / year (219.0M Units × ₹2.50)
Annual Contractual O&M + Insurance Expenses:	₹5.47 Crores / year (~10.0% of revenue)
Net Annual Pre-Tax EBITDA (P90 Operational):	₹49.28 Crores / year (90.0% EBITDA Margin)
Corporate Income Tax Rate (Section 115BAA):	25.168% (Base 22% + Surcharge 10% + Cess 4%)

The promoter evaluates two distinct financing options to construct this ₹500 Crore solar plant: - **Option A (Corporate Balance Sheet Term Loan):** Raising a standard corporate loan on the parent company's balance sheet at **11.50% p.a.** over a 6-year door-to-door tenure with an aggressive 6-month moratorium, requiring a **40% promoter cash equity contribution (₹200.00 Crores)**. - **Option B (Structured Non-Recourse SPV Solar Project Debt):** Raising a structured limited-recourse project finance facility inside the standalone bankruptcy-remote SPV at **8.75% p.a.** over an 18-year door-to-door tenure (18-month construction moratorium + 16.5-year sculpted post-COD amortization), requiring only a **25% promoter cash equity contribution (₹125.00 Crores)**.

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9.2 Financial Comparative Modeling: Option A vs. Option B

Option A: Corporate Balance Sheet Term Loan Model

- **Promoter Equity Contribution (40%): ₹200.00 Crores** (Heavy upfront equity lockup)
- **Corporate Term Loan Sized (60%):** ₹300.00 Crores
- **Interest Rate:** 11.50% p.a.
- **Moratorium:** 6 Months Only *(Forces immediate principal amortization during construction)*
- **Amortization Tenure:** 5.5 Years Post-Moratorium (₹54.55 Crores/year principal repayment)
- **Year 1 Debt Servicing Outflow (Construction Phase):**

$$extYear1Interest = ₹300.00extCrimes11.50\% = ₹34.50extCrores$$

$$extYear1PrincipalRepayment = ₹60.00extCrores$$

$$extTotalYear1Outflow = ₹94.50extCrores$$

- **Year 1 Project EBITDA: INR 0.00** *(Plant is under active construction; zero generation)*
- **Year 1 Cash Flow Deficit: -₹94.50 Crores (SEVERE CORPORATE LIQUIDITY DISTRESS / DEFAULT!)**
- **Year 2 Operational Debt Servicing Outflow (Post-COD):**

$$extInterestOutflow = ₹240.00extCrimes11.50\% = ₹27.60extCrores$$

$$extPrincipalOutflow = ₹54.55extCrores$$

$$extTotalDebtService = ₹82.15extCrores$$

- **Year 2 Operational DSCR (Option A):**

$$extDSCR_{extYear2} = racextCFADSextDebtService = rac₹49.28extCr₹82.15extCr = 0.599x \quad (extCRITICALCASHDEFAULT$$

Result under Option A: Sizing debt on a short corporate tenure forces annual debt service of ₹82.15 Crores against P90 EBITDA of only ₹49.28 Crores, resulting in a failing **0.599x DSCR**, triggering immediate loan default, SARFAESI enforcement, and corporate bankruptcy!

Option B: Structured Non-Recourse SPV Solar Project Debt Model

- **Promoter Equity Contribution (25%): ₹125.00 Crores**
- **Net Cash Equity Preserved vs Option A: +₹75.00 Crores** *(₹75.00 Crores in liquid equity saved)*
- **Senior SPV Solar Project Debt Sized (75%): ₹375.00 Crores**
- **Interest Rate: 8.75% p.a.** (Benchmark EBLR + 2.25% spread)
- **Moratorium Period: 18 Months** (Pre-COD Interest capitalized into CWIP under Ind AS 23)
- **Year 1 Cash Debt Servicing Outflow: INR 0.00** *(100% solvent; zero cash outlay during construction)*

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- **Amortization Tenure:** 16.5 Years Post-COD Sculpted Amortization (₹22.73 Crores/year average principal)
- **Year 2 Operational Debt Servicing Outflow (Stabilized COD):**

$$extYear2InterestOutflow = ₹375.00extCrimes8.75\% = ₹32.81extCrores$$

$$extYear2PrincipalOutflow(Sculpted) = ₹3.25extCrores$$

$$extTotalYear2DebtService = ₹36.06extCrores$$

- **Year 2 Operational DSCR (Option B):**

$$extDSCR_{extYear2} = racextCFADSextDebtService = rac₹49.28extCr₹36.06extCr = 1.367x \quad (extPRIMEINVESTMENTGRADE!)$$

Result under Option B: The 18-year sculpted amortization matches the project's P90 cash flow profile, achieving a robust **1.367x DSCR**, leaving an annual cash surplus of **₹13.22 Crores** in the TRA Escrow Account after servicing all debt!

9.3 Comprehensive Opportunity-Adjusted Wealth Surplus Synthesis

In addition to preventing a cash flow default, Option B unlocks massive financial wealth for the promoter through tax shields and equity capital preservation:

1. Section 32 Accelerated Depreciation Tax Shield Savings (Year 1):

$$extYear1SolarPlantWDVDepreciation = ₹500.00extCroresimes40\% = ₹200.00extCrores$$

$$extCorporateIncomeTaxSaved(25.168\%Rate) = ₹200.00extCrimes25.168\% = ₹50.336extCrores$$

2. Section 36(1)(iii) Interest Tax Shield Savings (5-Year Cumulative):

$$extCumulative5 - YearSPVInterestExpense = ₹142.68extCrores$$

$$extCorporateTaxShieldSaved(25.168\%Rate) = ₹142.68extCrimes25.168\% = ₹35.910extCrores$$

$$extTotalCumulativeTaxShieldSavings(Depreciation + Interest) = ₹86.246extCrores$$

3. Reinvestment Return on Preserved Equity Cash:

By contributing only ₹125 Crores in equity under Option B instead of ₹200 Crores under Option A, the promoter retains **₹75.00 Crores in liquid corporate cash**. Reinvested across the sponsor's corporate pipeline earning a conservative 16.0% p.a. Return on Capital Employed (ROCE) over 5 years:

$$ext5 - YearPost - TaxReinvestmentIncome = ₹75.00extCrimes[(1 + 0.16imes(1 - 0.25168))^5 - 1ight] =+ ₹45.00extCrores$$

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MSIPSL 100 MW SOLAR PROJECT - 5-YEAR FINANCIAL SURPLUS MATRIX		
Financial Performance Parameter	**Option A (Corporate)**	**Option B (Project Debt)**
Promoter Cash Equity Margin Required	₹200.00 Crores	₹125.00 Crores
Liquid Cash Equity Preserved for Sponsor	₹0.00	+₹75.00 Crores
Year 1 Construction Cash Deficit	-₹94.50 Crores (Default)	INR 0.00 (Solvent)
Year 2 Operational DSCR (P90 EBITDA)	0.599x (Failing)	1.367x (Prime Investment)
Nominal Interest Rate (% p.a.)	11.50% p.a.	8.75% p.a.
Effective Post-Tax Interest Cost	8.61% p.a.	6.55% p.a.
5-Year Section 32 & 36(1)(iii) Tax Shields	₹25.17 Crores	+₹86.25 Crores
5-Year Reinvestment Return on Preserved Equity	₹0.00	+₹45.00 Crores
Less Bank Upfront Setup Fees (1.0% / 0.75%)	-₹3.00 Crores	-₹2.81 Crores
NET STRATEGIC FINANCIAL WEALTH SURPLUS	**Baseline (Deficit)**	**+₹202.49 CRORES**

Strategic Outcome: By executing Option B (Structured Non-Recourse SPV Solar Project Debt), **MSIPSL generates a total opportunity-adjusted net financial wealth surplus of ₹202.49 Crores (+₹202.49 Crores) over Option A**, preserves **₹75.00 Crores in liquid corporate cash**, converts a catastrophic 0.599x default into a prime **1.367x DSCR**, unlocks **₹86.25 Crores in statutory tax shields**, and slashes its effective post-tax borrowing yield to **6.55% p.a.**!

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Module 10: Regulatory Compliance, RBI IRACP Norms & Default Framework

10.1 RBI Income Recognition, Asset Classification & Provisioning (IRACP) Directions

Solar project finance facilities are subject to strict regulatory oversight under the RBI IRACP Directions:

- **90-Day Overdue Rule:** If senior debt interest or principal installments remain unpaid for more than 90 consecutive days from the due date, the loan account is automatically classified as a **Non-Performing Asset (NPA)** under the Special Mention Account (SMA) framework:
  - *SMA-0:* Principal or interest overdue for 1 to 30 days.
  - *SMA-1:* Principal or interest overdue for 31 to 60 days.
  - *SMA-2:* Principal or interest overdue for 61 to 90 days.
  - *NPA (Sub-Standard):* Overdue exceeding 90 days.
- **RBI Standard Asset Provisioning Rates for Project Finance:**
  - *Construction Phase (Pre-COD):* Mandatory **5.0% standard asset provisioning** on outstanding principal.
  - *Post-COD Stabilized Phase:* Provisioning steps down to **2.5%** upon achieving Commercial Operations, and further decreases to **1.0%** after 2 years of operational cash flows achieving a minimum DSCR of $1.25x$.

10.2 Date of Commencement of Commercial Operations (DCCO) Monitoring & Extension Rules

Under RBI guidelines, every solar project sanction letter must explicitly define a target **Date of Commencement of Commercial Operations (DCCO)**.

Permissible DCCO Extensions Without NPA Downgrade:

1. **Infrastructure Solar Projects:** DCCO can be extended up to a maximum of **2 Years** from the original target date without triggering automatic NPA classification, provided the delay is caused by legal disputes, land acquisition bottlenecks, or grid transmission delays beyond the developer's control.
2. **Restructuring Framework:** The project loan can be restructured during the 2-year extended DCCO window while maintaining "Standard Asset" classification, provided the total repayment tenure is not extended beyond 85% of the remaining PPA economic life.

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10.3 Default Resolution, SARFAESI Enforcement & IBC Insolvency Mechanics

In the event of a sustained operational default or off-taker payment failure: 1. **SARFAESI Act Enforcement:** Lenders issue a 60-day statutory demand notice under **Section 13(2) of the SARFAESI Act, 2002**. If unfulfilled, lenders exercise power under **Section 13(4)** to take over physical possession of the solar park, seize the TRA Escrow Account, and auction the solar plant. 2. **PPA Step-in Rights Execution:** Lenders invoke the Tripartite Substitution Agreement with SECI/DISCOM, substituting the defaulting SPV promoter with an experienced operational solar O&M operator. 3. **Insolvency and Bankruptcy Code (IBC), 2016:** Senior lenders file an admission petition under **Section 7 of the IBC** before the National Company Law Tribunal (NCLT) to initiate Corporate Insolvency Resolution Process (CIRP) against the SPV.

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Module 11: Professional Certification Examination & Detailed Rationales

To verify complete conceptual, legal, technical, and mathematical mastery of Solar Project Funding in India, complete the following ten scenario-based examination questions. Detailed institutional answer rationales are provided immediately following the exam paper.

Certification Examination Paper

Question 1 (Solar Resource Assessment & Debt Sizing)

Why do institutional bank credit committees in India size solar project debt and schedule debt service strictly against P90 generation probability estimates rather than P50 estimates? - (A) P50 estimates reflect extreme 99% downside weather failure, which artificially inflates equity returns. - (B) P90 estimates represent a conservative generation figure expected to be equaled or exceeded 90% of the time, ensuring cash flow covers debt service even during lower-irradiation years. - (C) P50 estimates are legally barred by MNRE ALMM guidelines for utility-scale SECI projects. - (D) P90 estimates allow lenders to increase the debt-to-equity ratio beyond 90:10 without requiring a DSRA buffer.

Question 2 (Group Captive Equity & Consumption Mandate)

Under Rule 3 of the Electricity Rules 2005, what are the mandatory statutory equity and power consumption thresholds required for a Group Captive Solar Project to qualify for Cross-Subsidy Surcharge (CSS) exemption? - (A) Off-taker must hold at least 51% equity in the SPV and consume at least 26% of generated power. - (B) Off-taker must hold at least 26% equity in the SPV and consume at least 51% of generated power. - (C) Off-taker must hold 100% equity in the SPV and consume 100% of generated power. - (D) Off-taker must hold 10% equity in the SPV and consume 75% of generated power.

Question 3 (Technical CUF Calculation)

Calculate the annual Capacity Utilization Factor (CUF %) of a 100 MW AC / 120 MW DC solar PV plant that generates 210.24 Million Units (kWh) of electricity over a 365-day year (8,760 hours): - (A) 20.00% - (B) 24.00% - (C) 28.00% - (D) 32.00%

Question 4 (Section 32 Accelerated Depreciation Tax Shield)

A corporate sponsor investing ₹100 Crores in a Group Captive Solar plant is subject to a corporate tax rate of 25.168%. Calculate the direct Year 1 corporate tax cash shield saved under Section 32 (40% WDV accelerated depreciation) if put to use for over 180 days: - (A) ₹4.00 Crores - (B) ₹10.067 Crores - (C) ₹15.00 Crores - (D) ₹25.168 Crores

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Question 5 (TRA Escrow Priority Waterfall)

In a 7-tier Trust and Retention Account (TRA) Escrow Cash Flow Waterfall, where does Senior Debt Principal Repayment sit relative to Essential O&M Expenses and DSRA Reserve Top-Ups? - (A) Above Essential O&M Expenses and below DSRA Reserve Top-Ups. - (B) Below Essential O&M Expenses and Senior Debt Interest, but above DSRA Reserve Top-Ups. - (C) Below Subordinated Debt Servicing and above Statutory Taxes. - (D) At the very bottom below Free Surplus Cash Distribution to Sponsor Equity.

Question 6 (PPA Payment Security Mechanism)

What primary Payment Security Mechanism (PSM) does SECI provide under utility-scale 25-year PPAs to mitigate state DISCOM payment default risk? - (A) An unsecured corporate guarantee issued by the State PWD Department. - (B) A 3-month revolving Letter of Credit (LC) backed by a Payment Security Fund and RBI Tripartite Agreement. - (C) An equity pledge over 100% of DISCOM shares held by the Ministry of Finance. - (D) A mandatory 50% cash margin deposited in the SPV's current account.

Question 7 (Statutory Charge Registration Deadline)

Under Section 77 of the Companies Act 2013, within how many days from execution must an exclusive corporate charge over a solar SPV's assets be filed with the Registrar of Companies (ROC Form CHG-1)? - (A) 15 Days - (B) 30 Days - (C) 60 Days - (D) 90 Days

Question 8 (Effective Post-Tax Debt Yield Calculation)

If a solar SPV borrows at a nominal floating interest rate of 9.00% p.a. and is subject to the corporate tax rate of 25.168%, what is its effective post-tax borrowing yield under Section 36(1)(iii)? - (A) 9.00% p.a. - (B) 7.50% p.a. - (C) 6.735% p.a. - (D) 5.25% p.a.

Question 9 (RBI DCCO Project Extension Rules)

Under RBI guidelines, up to what maximum duration can the target Commercial Operation Date (DCCO) of an infrastructure solar project be extended without triggering automatic NPA classification? - (A) 6 Months - (B) 1 Year - (C) 2 Years - (D) 5 Years

Question 10 (Debt Service Coverage Ratio Calculation)

Calculate the Year 2 operational Debt Service Coverage Ratio (DSCR) for a solar SPV with annual CFADS of ₹40.00 Crores, scheduled senior debt interest of ₹20.00 Crores, and scheduled principal repayment of ₹12.00 Crores: - (A) 1.00x - (B) 1.25x - (C) 1.50x - (D) 2.00x

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Comprehensive Certification Answer Keys & Analytical Rationales

Answer 1: (B)

Rationale: P90 generation estimates represent a conservative generation level expected to be equaled or exceeded 90% of the time based on 20-year solar irradiation statistics. Sizing debt strictly against P90 guarantees that even during lower-irradiation or cloudy years, project CFADS will comfortably service 100% of senior debt principal and interest without defaulting.

Answer 2: (B)

Rationale: Under Rule 3 of the Electricity Rules 2005, a Group Captive power structure legally requires the captive off-taker(s) to hold at least **26% of the equity shareholding** in the SPV and consume at least **51% of the total electricity generated**. Fulfilling both conditions grants full statutory exemption from Cross-Subsidy Surcharges (CSS) and Additional Surcharges.

Answer 3: (B)

Rationale: CUF is calculated relative to AC nameplate capacity:

$$extCUF(\%) = rac210,240,000extkWh100,000extkWimes8,760extHoursimes100 = rac210,2$$

Answer 4: (B)

Rationale: Year 1 tax depreciation under Section 32 = $₹100.00extCroresimes40\% = ₹40.00extCrores$. Applying the corporate tax rate of 25.168%:

$$extTaxCashSaved = ₹40.00extCroresimes25.168\% = ₹10.0672extCrores$$

Answer 5: (B)

Rationale: Under the 7-tier TRA Escrow Waterfall, Tier 1 is Statutory Taxes, Tier 2 is Essential O&M Expenses, Tier 3 is Senior Debt Interest, Tier 4 is Senior Debt Principal Repayment, Tier 5 is Reserve Top-Ups (DSRA/IRRA), Tier 6 is Subordinated Debt, and Tier 7 is Free Surplus to Sponsor Equity. Thus, Senior Principal sits below O&M/Interest and above Reserve Top-Ups.

Answer 6: (B)

Rationale: SECI utility PPAs incorporate a multi-layered Payment Security Mechanism (PSM) consisting of a 3-month revolving Letter of Credit (LC) opened by SECI in favor of the SPV, backed by SECI's Payment Security Fund and an RBI Tripartite Agreement (TPA) with State Governments.

Answer 7: (B)

Rationale: Section 77(1) of the Companies Act 2013 mandates that every company creating a charge on its property or assets must file the charge details in ROC Form CHG-1 with the Registrar of Companies within **30 days** of charge creation.

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Answer 8: (C)

Rationale: Under Section 36(1)(iii), interest paid on business debt is fully tax-deductible:

$$ ext{Effective Yield} = 9.00\% imes (1 - 0.25168) = 9.00\% imes 0.74832 = \mathbf{6.73488\% \quad (pprox 6.735\% ext{ p.a.})}$$

Answer 9: (C)

Rationale: RBI Project Finance Directions permit the DCCO of infrastructure projects (including solar farms) to be extended up to a maximum of **2 Years** from the original target COD due to court disputes, land acquisition bottlenecks, or grid transmission delays without forcing automatic NPA classification.

Answer 10: (B)

Rationale: Debt Service = Scheduled Interest (₹20.00 Cr) + Scheduled Principal (₹12.00 Cr) = ₹32.00 Crores.

$$extDSCR = racextCFADSextDebtService = rac₹40.00extCrores₹32.00extCrores = 1.25x$$

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